

TFM - μy Vs. y : TT plots

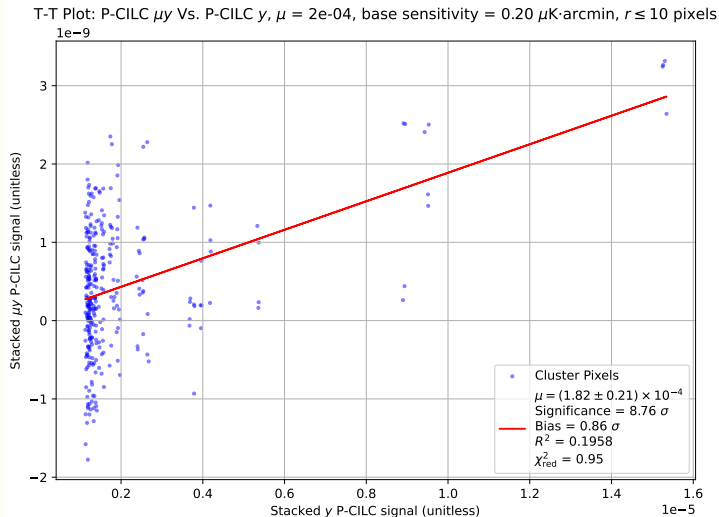
Joan Ribot Oliver

2026

Universidad de Cantabria

TT plot: P-CILC

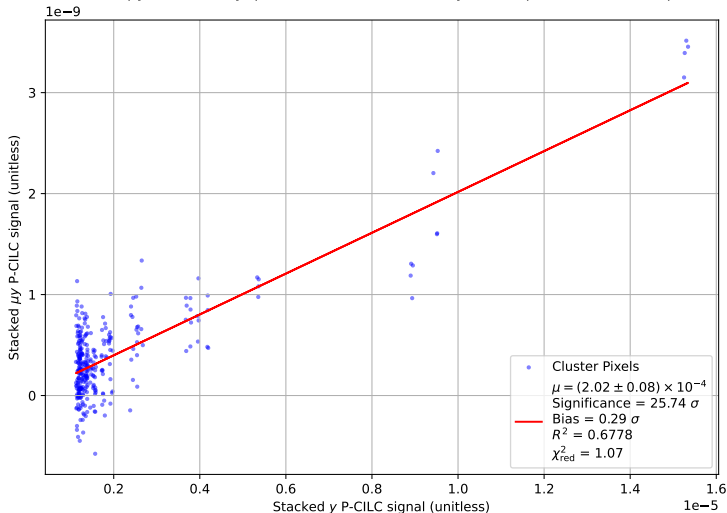
4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4
 $\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-CILC

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4
 $\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$

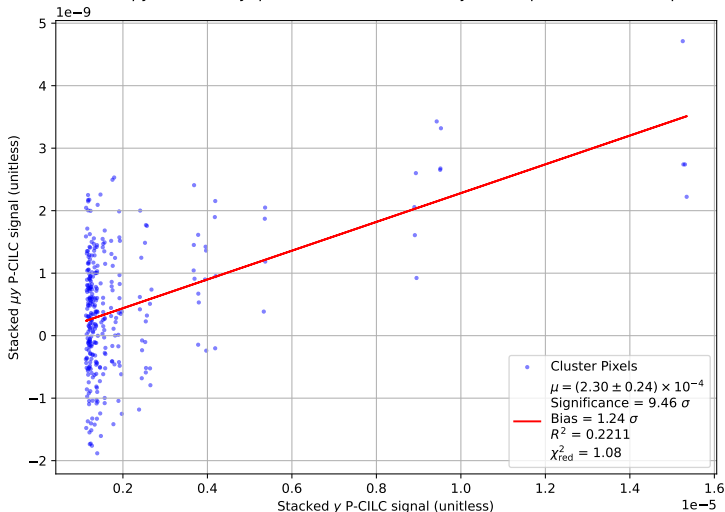
T-T Plot: P-CILC μy Vs. P-CILC y , $\mu = 2\text{e-}04$, base sensitivity = $0.20 \mu K \cdot \text{arcmin}$, $r \leq 10$ pixels, PICO



TT plot: P-CILC

4 layers: tSZ, μ -distortion, CMB and noise ($0.61 \mu K \cdot \text{arcmin}$), s.m. = 10^4
 $\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$

T-T Plot: P-CILC μy Vs. P-CILC y , $\mu = 2\text{e-}04$, base sensitivity = $0.61 \mu K \cdot \text{arcmin}$, $r \leq 10$ pixels, PICO

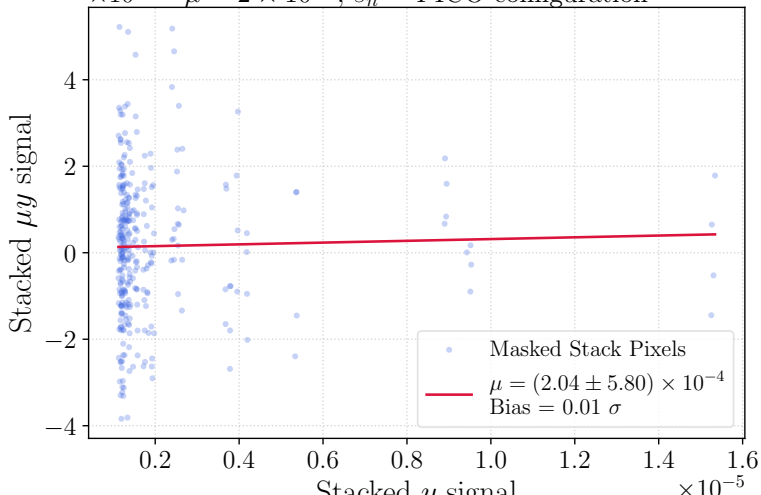


TT plot: P-CILC

4 layers: tSZ, μ -distortion, CMB and noise (PICO), s.m. = 10^4

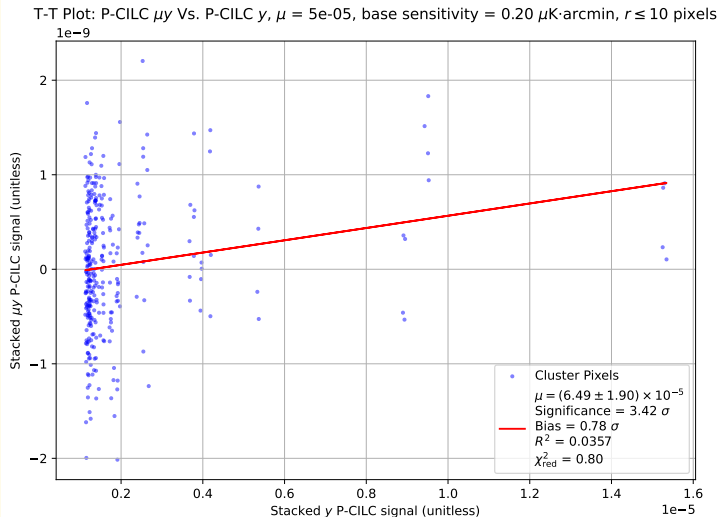
$\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$

T-T Plot for μ extraction, $N = 10^4$, 200 iterations, PICO
 $\times 10^{-8}$ $\mu = 2 \times 10^{-4}$, s_n = PICO configuration



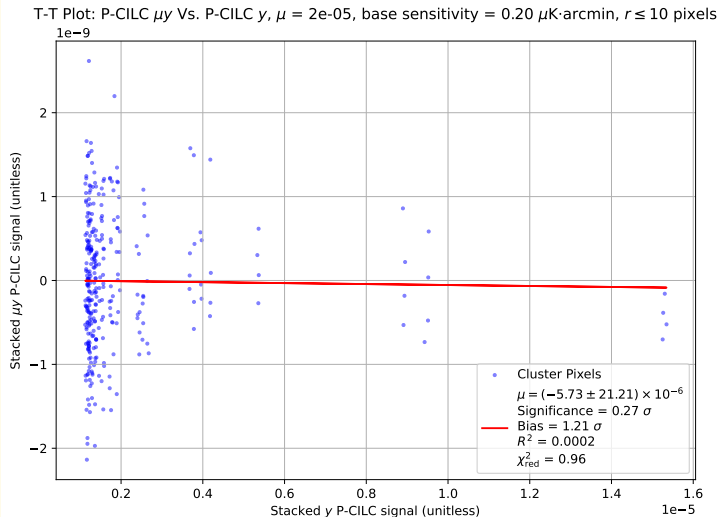
TT plot: P-CILC

4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4
 $\mu = 5 \times 10^{-5}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-CILC

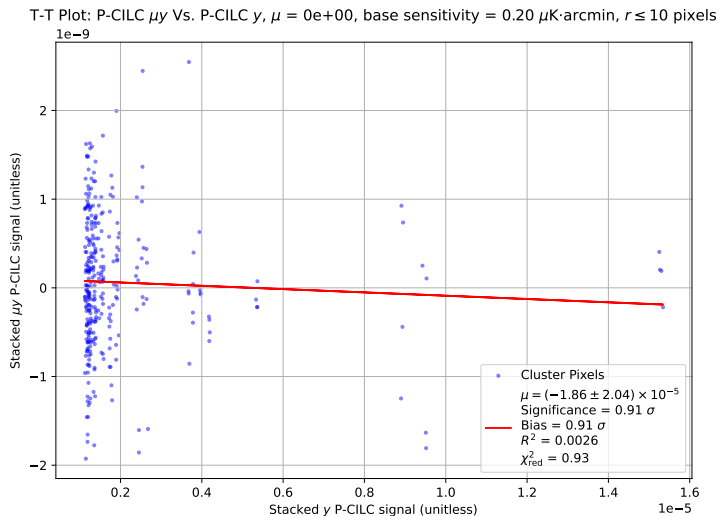
4 layers: tSZ, μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4
 $\mu = 2 \times 10^{-5}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-CILC

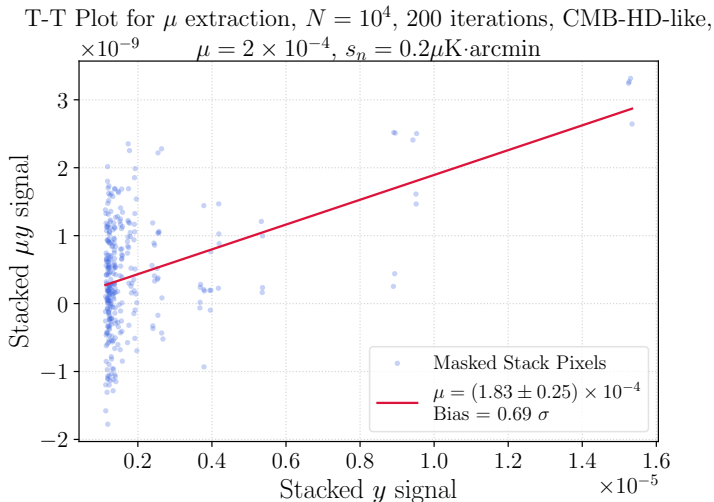
3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4

Null test: $\mu = 0$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



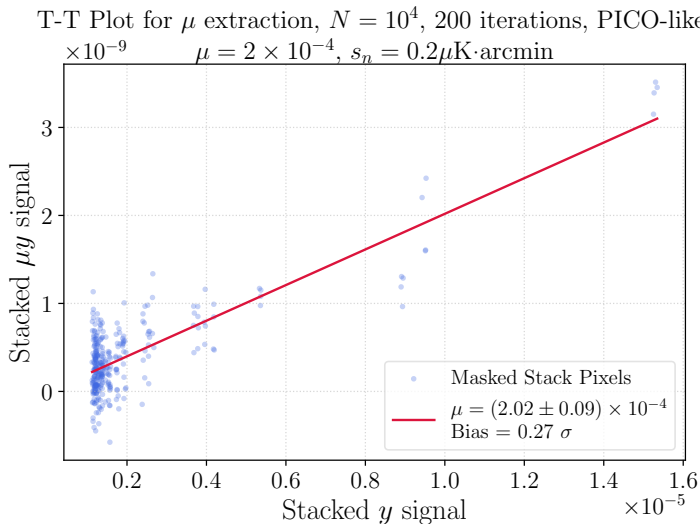
TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$), s.m. = 10^4 , Bootstrap
 $\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



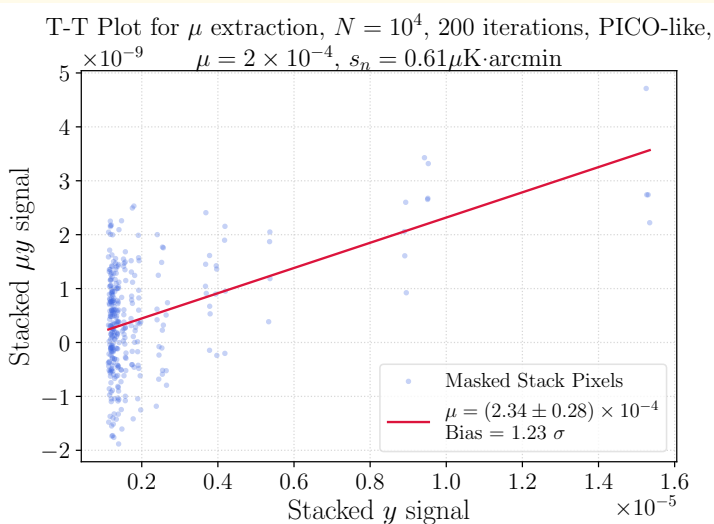
TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ($0.2\mu\text{K}\cdot\text{arcmin}$), s.m. = 10^4 , Bootstrap
 $\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$



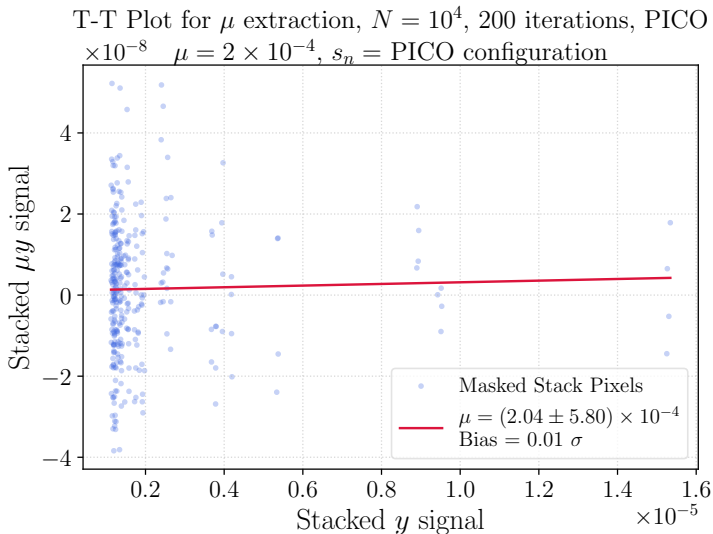
TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ($0.61 \mu\text{K} \cdot \text{arcmin}$), s.m. = 10^4 , Bootstrap
 $\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$



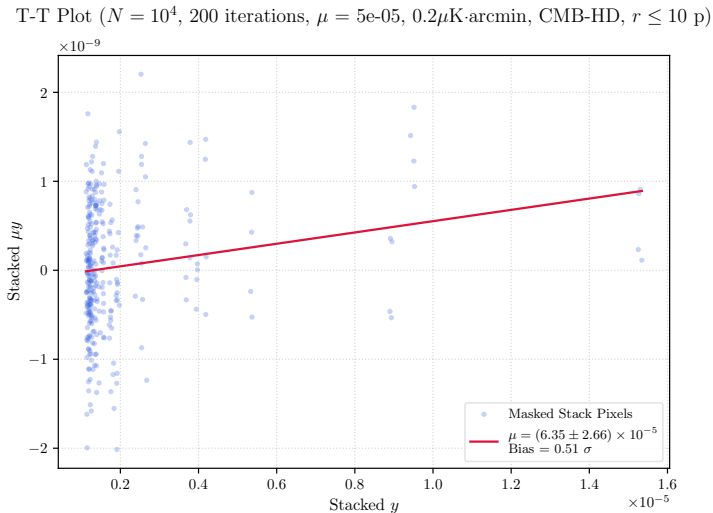
TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise (PICO), s.m. = 10^4 , Bootstrap
 $\mu = 2 \times 10^{-4}$, freqs = PICO, $r = 10$ pixels = $15.0'$



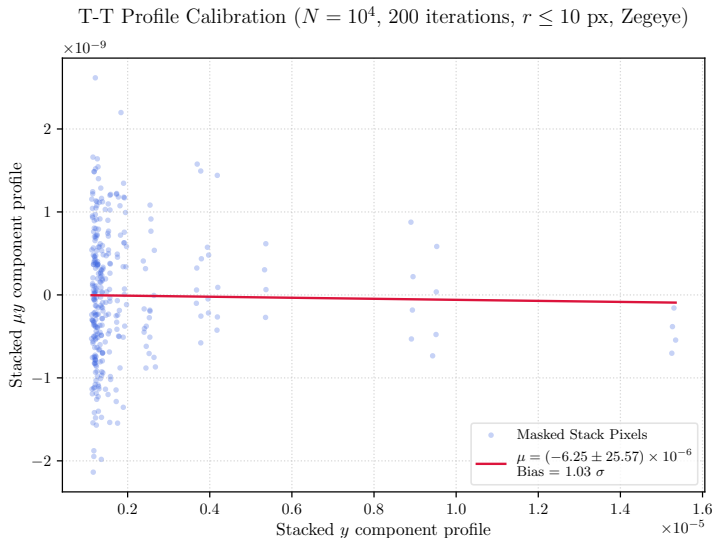
TT plot: P-CILC + BOOTSTRAP

4 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4 , Bootstrap
 $\mu = 5 \times 10^{-5}$, freqs = *CMB-HD*, $r = 10$ pixels = $15.0'$



TT plot: P-CILC + BOOTSTRAP

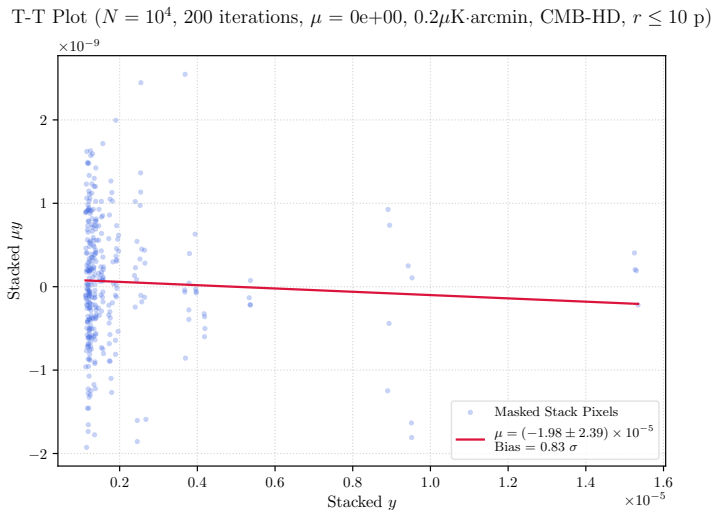
4 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4 , Bootstrap
 $\mu = 2 \times 10^{-5}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-CILC + BOOTSTRAP

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4 , Bootstrap

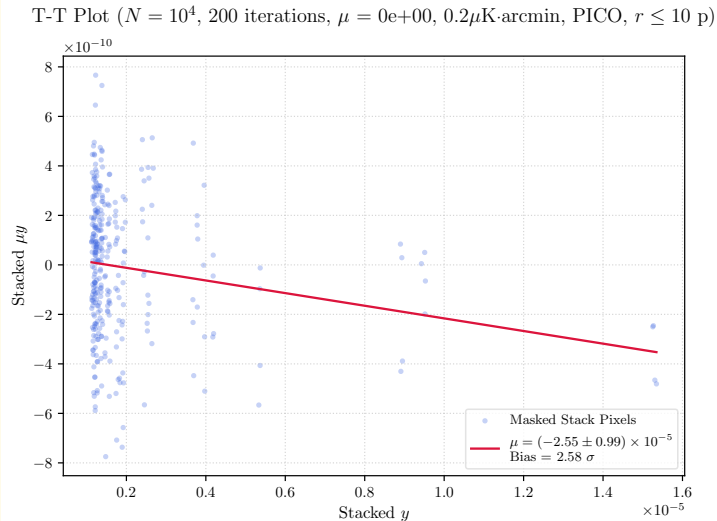
Null test: $\mu = 0$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-CILC + BOOTSTRAP

3 layers: tSZ, CMB and noise ($0.2\mu K \cdot \text{arcmin}$), s.m. = 10^4 , Bootstrap

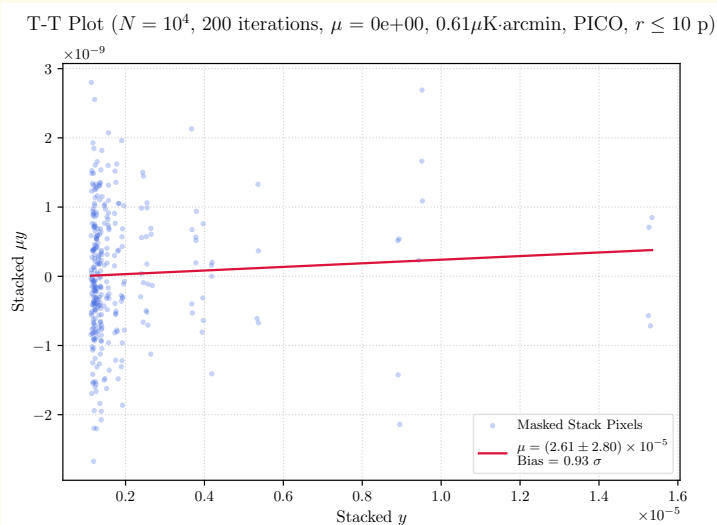
Null test: $\mu = 0$, freqs = *PICO*, $r = 10$ pixels = $15.0'$



TT plot: P-CILC + BOOTSTRAP

3 layers: tSZ, CMB and noise ($0.61\mu\text{K}\cdot\text{arcmin}$), s.m. = 10^4 , Bootstrap

Null test: $\mu = 0$, freqs = *PICO*, $r = 10$ pixels = $15.0'$

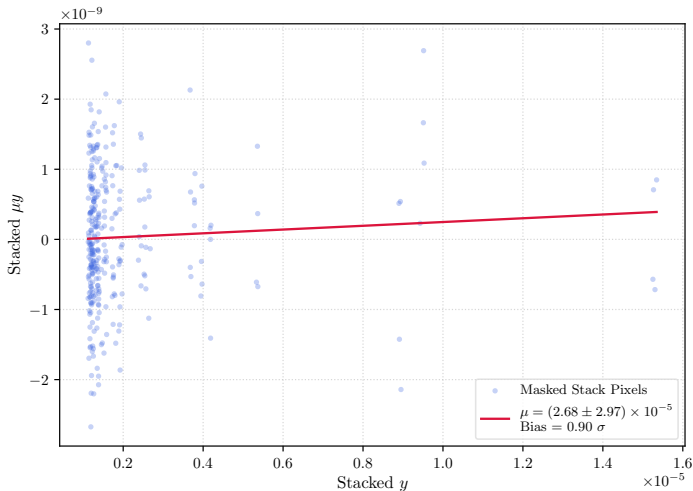


TT plot: P-CILC + BOOTSTRAP

3 layers: tSZ, CMB and noise (PICO), s.m. = 10^4 , Bootstrap

Null test: $\mu = 0$, freqs = *PICO*, $r = 10$ pixels = $15.0'$

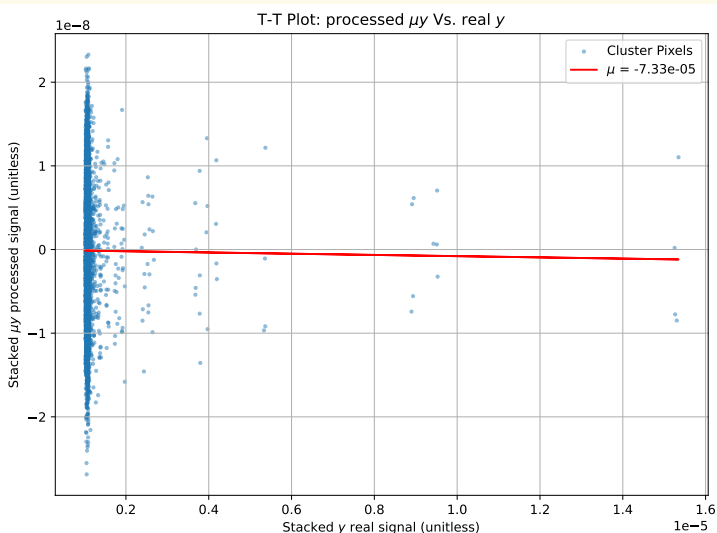
T-T Plot ($N = 10^4$, 200 iterations, $\mu = 0\text{e}+00$, PICO configuration, $r \leq 10$ p)



TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($30\mu K \cdot \text{arcmin}$)

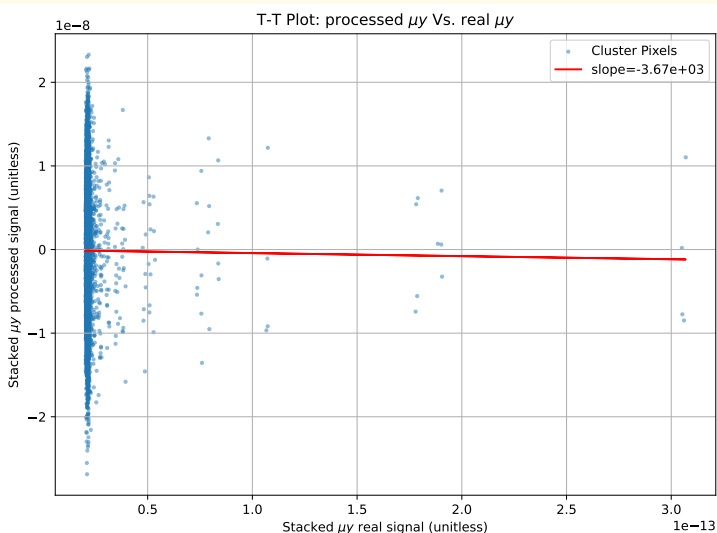
$$\mu = 2 \times 10^{-8}, \text{ freqs} = \text{CMB-HD}$$



TT plot: P-ILC no γ input

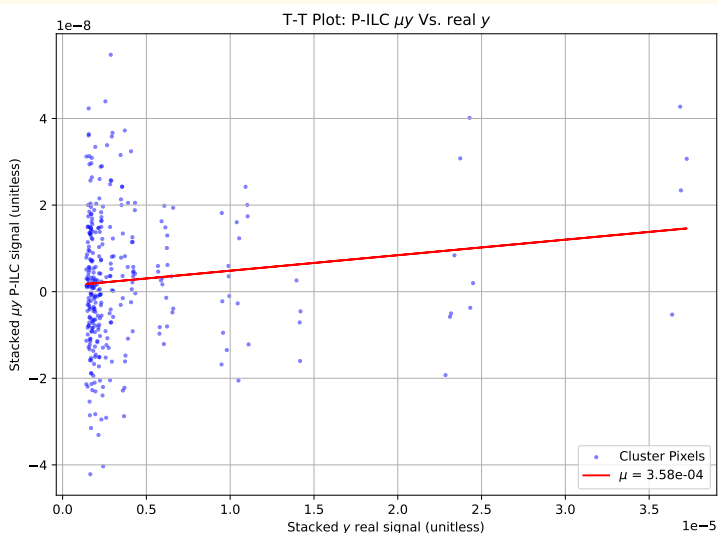
3 layers: μ -distortion, CMB and noise ($30\mu K \cdot \text{arcmin}$)

$$\mu = 2 \times 10^{-8}, \text{ freqs} = \text{CMB-HD}$$



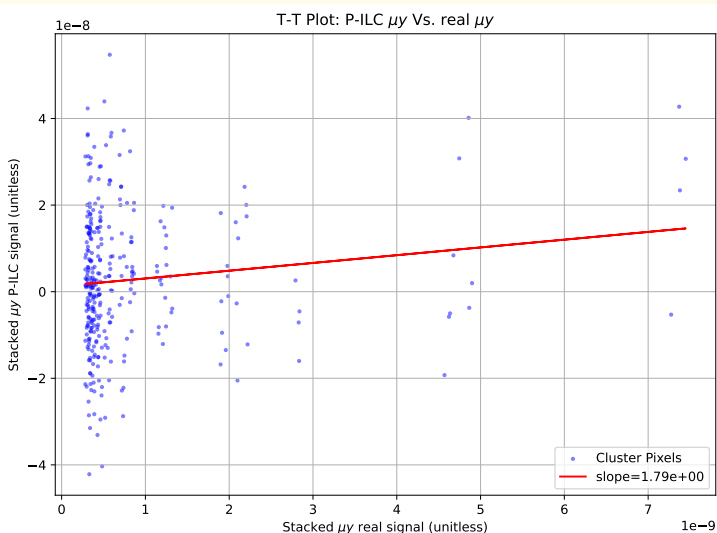
TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-ILC no y input

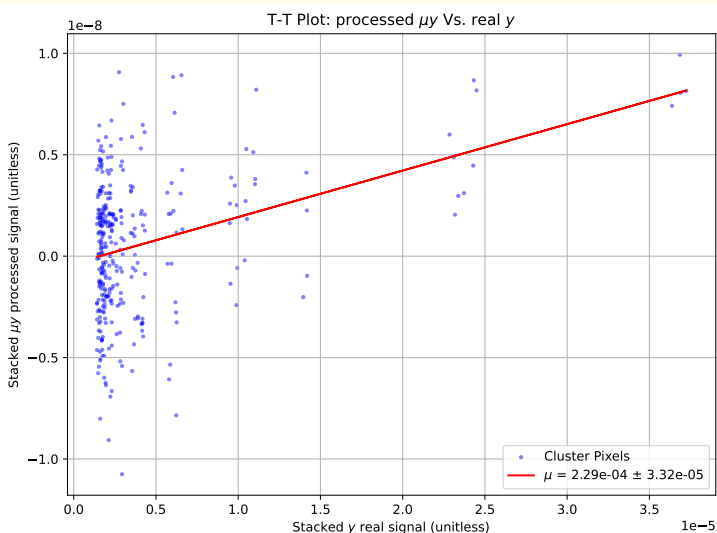
3 layers: μ -distortion, CMB and noise ($2\mu K \cdot \text{arcmin}$)
 $\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

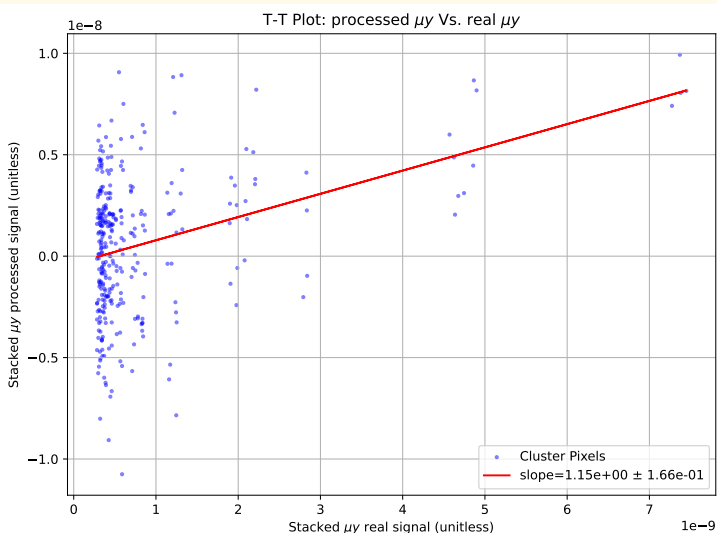
$\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($0.2\mu K \cdot \text{arcmin}$)

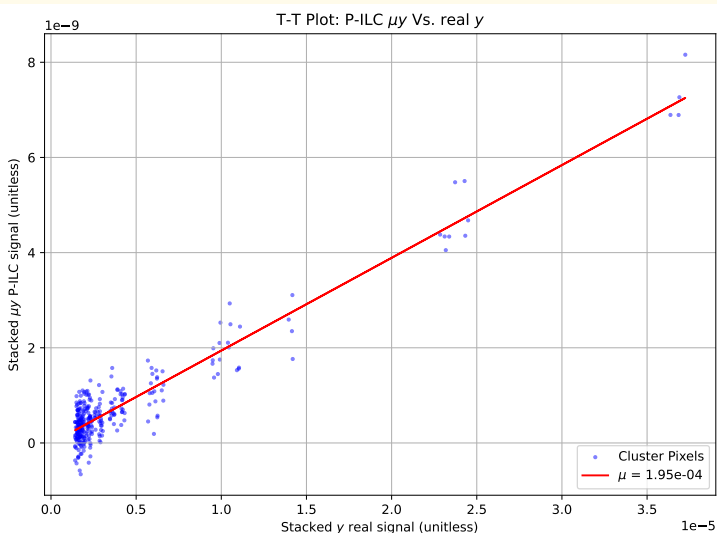
$\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($0.02\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$



TT plot: P-ILC no y input

3 layers: μ -distortion, CMB and noise ($0.02\mu K \cdot \text{arcmin}$)

$\mu = 2 \times 10^{-4}$, freqs = CMB-HD, $r = 10$ pixels = $15.0'$

